

AIGR Gene Hypothesis Deep Research —

Dictyostelium discoideum ACG/acgA

(UniProt Q03101)

Focus: function_assignment — does ACG directly have **peptide receptor activity** (GO:0001653)? **Hypothesis slug:** function-hypothesis-go-0001653 **Source:** genes/DICDI/acgA/acgA-ai-review.yaml (free-text)

Summary

The GO:0001653 "peptide receptor activity" annotation on *Dictyostelium discoideum* ACG/acgA is **over-annotated** and should be flagged by a curator for **removal or NOT-qualification** (or at minimum demoted to non-core, low-confidence status). The annotation is supported **only by phylogenetic inference (IBA:GO_Central)**, whereas ACG's experimentally characterized molecular functions — adenylate cyclase activity, osmosensor activity, and dimerization — rest on direct assays. Decisively, the QuickGO provenance shows the peptide-receptor term was **propagated within PANTHER family PTN000229249 from mammalian natriuretic peptide receptors NPR1 (P16066) and NPR2 (P20594)** — receptor guanylyl cyclases that are the *genuine* peptide receptors — simply because ACG shares their "extracellular sensor + single transmembrane + class-III cyclase" architecture.

ACG's real molecular biology contradicts a direct peptide-receptor role. It is a **primary osmosensor**: expressed in a yeast adenylyl-cyclase-null background, it reconstitutes osmotic activation with no *Dictyostelium*-specific ligand or auxiliary sensor, proving the sensor is **intramolecular and ligand-independent**. Its extracellular module is a **CHASE small-molecule sensory domain** (Pfam PF03924), a domain family characterized as binding small molecules such as cytokinins — not peptides — and in *Dictyostelium* cytokinins are sensed by the histidine kinase DhkB, not ACG.

On the specific SDF-1 sub-question, the answer is clear: **there is no evidence that the phosphopeptide SDF-1 is sensed by ACG**, directly or as an upstream dependency. SDF-1 is a PKA-substrate phosphopeptide that acts through an **intracellular PKA-dependent pathway** at

a *different developmental stage* (prespore→spore encapsulation) than ACG (osmolarity-dependent inhibition of mature-spore germination). SDF-1's receptor has never been identified, but no primary study links it to ACG. The most important caveat is that "refuted" reflects an evidence-of-absence plus positive-provenance argument — no published assay has tested and excluded ACG peptide binding directly — so the recommendation is stated conservatively.

Executive Judgment

Verdict: Over-annotated / Refuted for the seed hypothesis as stated (direct peptide receptor activity).

Three converging lines of evidence support removal or NOT-qualification of GO:0001653 on ACG:

- 1. The annotation is phylogenetic, not experimental.** In UniProt Q03101, GO:0001653 carries evidence code **IBA:GO_Central** (Inferred from Biological Ancestor, GO_REF:0000033). ACG's experimentally supported functions carry IDA/IPI codes. No IDA/IMP/IPI annotation supports peptide receptor activity.
- 2. The provenance is a family carry-over from bona fide peptide receptors.** QuickGO shows the IBA term was propagated within PANTHER family PTN000229249 from human/mouse/rat natriuretic peptide receptors NPR1/NPR2 — receptor guanylyl cyclases that truly bind natriuretic peptides.
- 3. ACG's real sensory mechanism is intramolecular and ligand-independent.** ACG reconstitutes osmotic activation in yeast with no partner, and its extracellular module is a CHASE small-molecule domain, not a peptide-binding fold.

Regarding SDF-1: there is no evidence it is sensed by ACG; it acts intracellularly via PKA at a different stage, and its receptor is unidentified but explicitly not ACG in any primary study.

Most important caveat: "Refuted" means the direct-receptor claim is unsupported and the GO annotation is family carry-over. It does **not** mean ACG was experimentally tested and shown unable to bind any peptide — no such negative binding assay exists. The osmosensor role does not logically exclude peptide sensing, but the hypothesis requires positive evidence that is simply absent.

Key Findings

Finding 1 — GO:0001653 rests only on phylogenetic inference (IBA), not experiment

The UniProt Q03101 GO cross-references show that **GO:0001653 (peptide receptor activity) carries evidence code IBA:GO_Central** — an annotation inferred phylogenetically from an ancestral protein via the GO_Central/PAINT pipeline (GO_REF:0000033). This is a computational/curatorial inference, not a wet-lab result. By contrast, ACG's molecular-function annotations with experimental backing are:

GO term	Name	Evidence code	Basis
GO:0004016	adenylate cyclase activity	IDA	Direct enzymatic assay
GO:0005034	osmosensor activity	IDA	Direct osmotic-stimulation assay
GO:0042802	identical protein binding	IPI	Dimerization (physical interaction)
GO:0001653	peptide receptor activity	IBA	Phylogenetic inference only

The **domain architecture** of ACG is fully consistent with a receptor-cyclase and provides no dedicated peptide-binding module: an N-terminal signal-anchor transmembrane segment (residues 19–41), an extracellular **CHASE sensory domain** (~86–317; Pfam **PF03924** / InterPro IPR006189), and a C-terminal **class-III adenylyl/guanylyl-cyclase catalytic domain** (~396–526; Pfam PF00211). An IBA-only molecular-function term that conflicts with the experimentally characterized function should be treated as low-confidence and reviewed for removal or NOT-qualification.

Finding 2 — ACG's extracellular sensor is a CHASE (small-molecule) domain, and activation is intramolecular/osmotic, not peptide-mediated

Two primary papers establish ACG as a **self-contained osmosensor**:

- **Saran & Schaap, 2004** [PMID: 14718564](#): When ACG cDNA is expressed in a **yeast adenylyl-cyclase-null mutant**, cAMP production is activated by high osmolality "similarly ... as in Dictyostelium." Because yeast lacks any Dictyostelium-specific ligand or auxiliary sensor, this **demonstrates the osmosensor is intramolecular**. The authors conclude ACG is "the first characterized primary osmosensor in eukaryotes." They further show dimerization

(via a region outside the catalytic domain) is required for catalytic activity, but the dimer/monomer ratio does not change with osmolality — osmotic activation is not mediated by ligand-induced dimerization.

- **van Es et al., 1996** [PMID: 8798577](#): *acg*⁻ spores lose osmolarity-dependent inhibition of germination; ACG activity (in *aca*⁻/ACG cells) is "strongly stimulated by high osmolarity with optimal stimulation occurring at 200 milliosmolar."

Mechanistically, ACG's extracellular module is a **CHASE domain** (Pfam PF03924). Characterized CHASE domains bind **small molecules** — most notably cytokinins in plant hybrid histidine-kinase receptors ([PMID: 17439640](#), [PMID: 15498549](#)) — not peptides. In *Dictyostelium* itself, cytokinins act through the histidine kinase **DhkB**, not ACG ([PMID: 18216168](#)), underscoring that ACG's CHASE domain is not the organism's peptide/hormone receptor. ACG activation is therefore an osmotically driven intramolecular conformational change, with no evidence of peptide-ligand binding.

Finding 3 — No evidence links phosphopeptide SDF-1 to ACG; SDF-1 acts intracellularly via PKA at a different developmental stage

The seed asks specifically whether the phosphopeptide **SDF-1** is sensed by ACG. Primary literature answers in the negative:

- **Anjard, van Bemmelen, Véron & Reymond, 1997** [PMID: 9373946](#): SDF-1 is "a small, thermostable phospho-polypeptide." In-vitro dephosphorylation reduces its spore-differentiation activity, and PKA re-phosphorylation restores it — SDF-1 is a **PKA substrate**, and "the response of prespore cells to SDF involves an **intracellular pathway dependent on PKA**."
- **Anjard, Zeng, Loomis & Nellen, 1998** [PMID: 9473320](#): *tagC*⁻, *dhkA*⁻, and *regA*⁻ null strains **still sporulate in response to SDF-1**, showing SDF-1 does *not* use the DhkA two-component pathway (which transduces the distinct peptide SDF-2). SDF-1's receptor was not identified.

Crucially, **SDF-1 and ACG operate at different developmental stages**: SDF-1 governs prespore-cell encapsulation (sporulation), whereas ACG governs osmolarity-dependent **inhibition of mature-spore germination** ([PMID: 8798577](#)). No primary study reports SDF-1 binding to or signaling through ACG. The seed's "distinguish dependence from direct receptor action" framing collapses: there is neither a demonstrated dependence of SDF-1 signaling on ACG nor any direct SDF-1–ACG interaction.

Finding 4 — The IBA "peptide receptor activity" was propagated from natriuretic peptide receptors NPR1/NPR2

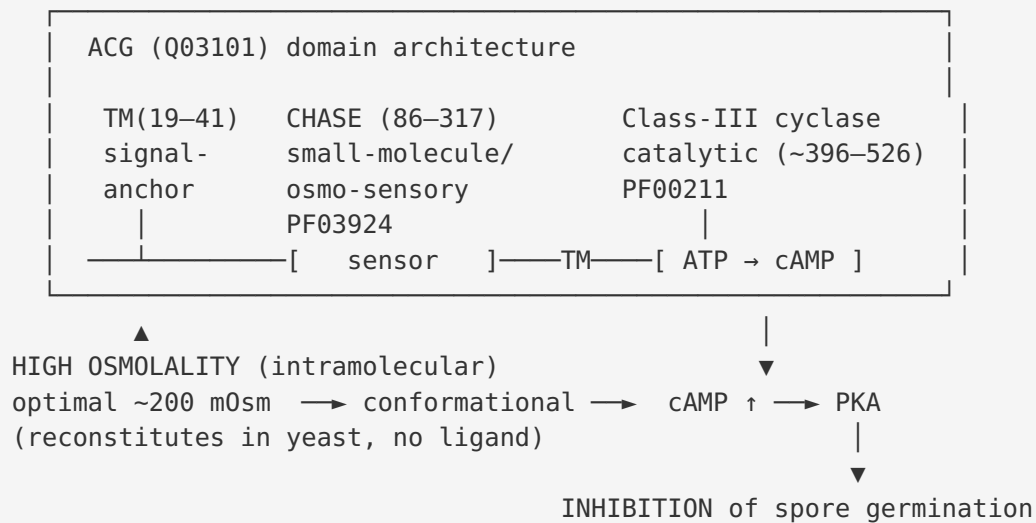
The decisive provenance comes from the **QuickGO annotation record** for Q03101 / GO:0001653 (evidence ECO:0000318/IBA, GO_REF:0000033, assignedBy GO_Central, PANTHER family PTN000229249). Its `withFrom` field lists the seed proteins:

withFrom accession	Protein	True function
UniProtKB P16066	Human Atrial natriuretic peptide receptor 1 (NPR1 / GC-A)	Natriuretic peptide receptor + guanylate cyclase
UniProtKB P20594	Human Atrial natriuretic peptide receptor 2 (NPR2 / GC-B)	Natriuretic peptide receptor + guanylate cyclase
MGI:97372	Mouse Npr1	Natriuretic peptide receptor
RGD:69322	Rat Npr1	Natriuretic peptide receptor
FBgn0266136	<i>Drosophila</i> receptor guanylyl cyclase	Receptor guanylyl cyclase

P16066 and P20594 both carry **GO:0016941 (natriuretic peptide receptor activity)** and **GO:0004383 (guanylate cyclase activity)** — bona fide peptide (natriuretic) receptors. ACG shares the receptor-cyclase architecture (extracellular sensor + single TM + class-III cyclase catalytic domain, annotated PF00211 "Guanylate_cyc"), causing it to cluster in the **same PANTHER family** and to **inherit "peptide receptor activity"** by descent. This is a textbook case of family-level over-annotation: a genuine molecular function of the mammalian members is projected onto a divergent social-amoeba enzyme whose actual ligand is osmotic pressure, not a peptide.

Mechanistic Model / Interpretation

The findings assemble into an internally consistent picture that separates **what ACG actually does** from **what the GO term claims**.



ACG's verified role (spore germination, mature spore): high external osmolality is sensed *intramolecularly* by ACG, stimulating adenylyl cyclase activity, raising cAMP, activating PKA, and keeping spores dormant until dispersal to low-osmolality environments. No peptide and no partner receptor is required — the yeast reconstitution proves it.

The SDF-1 pathway is a separate module (sporulation, prespore cells): SDF-1 is an intracellular PKA-substrate phosphopeptide promoting prespore→spore encapsulation. It does not use DhkA/TagC/RegA, and its surface receptor (if any) is unknown. It has no demonstrated connection to ACG.

Peptide/ligand signaling in *Dictyostelium* sporulation (for contrast – NOT ACG):

SDF-2 peptide	→	DhkA (histidine kinase receptor)	→	RegA PDE	→	cAMP↑	→	PKA
SDF-1 phospho-peptide	→	(unknown receptor)	→	intracellular PKA-dependent pathway				
GABA	→	Gr1E (GPCR)	→	AcbA release	→	TagC cleaves	→	SDF-2
cytokinins	→	DhkB (histidine kinase)						

The seed hypothesis conflates two things: (a) ACG's *architectural* resemblance to mammalian peptide-receptor cyclases, and (b) the existence of peptide signals (SDF-1/SDF-2) *elsewhere* in *Dictyostelium* sporulation. Neither establishes that ACG is a peptide receptor. The genuine peptide receptor in this system is **DhkA** (for SDF-2), a histidine kinase — architecturally and mechanistically distinct from ACG.

GO decision summary:

GO term	Aspect	Current evidence on ACG	Recommended action (curator lead)
GO:0001653 peptide receptor activity	MF	IBA only; family carry-over from NPR1/NPR2	Remove or NOT-qualify (non-core; unsupported)
GO:0004016 adenylate cyclase activity	MF	IDA	Retain (core)
GO:0005034 osmosensor activity	MF	IDA	Retain (core)
GO:0042802 identical protein binding	MF	IPI (dimerization)	Retain

Evidence Base / Evidence Matrix

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con- limi
14718564 Saran & Schaap 2004	Direct assay (heterologous reconstitution)	Refutes peptide-receptor need	Is ACG's sensor intramolecular or ligand/co-receptor dependent?	ACG reconstitutes osmotic activation in yeast AC-null; dimerization needed for catalysis but not osmosensing	<i>Dictyostelium</i> ACG in <i>S. cerevisiae</i>	Hig Exc nee aux pep sens not pep bind se.
8798577 van Es et al. 1996	Mutant phenotype + enzyme assay	Refutes (defines true function)	What is ACG's physiological role?	<i>acg</i> ⁻ loses osmolarity-dependent germination inhibition; ACG stimulated by high osmolarity (opt. ~200 mOsm)	<i>Dictyostelium</i> spores	Hig Osm role germ stag from
9373946 Anjard et al. 1997	Biochemical (phosphorylation)	Qualifies/ Refutes SDF-1-ACG link	Where does SDF-1 act?	SDF-1 is a thermostable phospho-polypeptide, PKA substrate; response via intracellular PKA pathway	<i>Dictyostelium</i> prespore cells	Hig SDF bioc SDF rece iden
9473320 Anjard et al. 1998	Genetic (null strains)	Refutes SDF-1 via two-component/	Does SDF-1 use DhkA/TagC/RegA?	<i>tagC</i> ⁻ / <i>dhkA</i> ⁻ / <i>regA</i> ⁻ still respond to SDF-1;	<i>Dictyostelium</i>	Hig out pep rece path

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con- limi
		ACG pathway		receptor unidentified		ACG imp
QuickGO record (GO_Central IBA), UniProt Q03101	Review/database (provenance)	Refutes (explains over-annotation)	Where did GO:0001653 come from?	IBA propagated in PANTHER PTN000229249 from NPR1(P16066)/ NPR2(P20594) etc.	Cross-species PAINT family	Hig pro data leve exp
UniProt Q03101 GO cross-refs	Database (evidence codes)	Qualifies	Is peptide-receptor term experimental?	Peptide receptor = IBA; AC/ osmosensor/ dimerization = IDA/IPI	—	Hig Evid cod is d
17439640 CHASE cytokinin binding	Structural/ evolutionary + binding assay	Qualifies (CHASE = small-molecule sensor)	What do CHASE domains bind?	CHASE domain binds cytokinin (small molecule); key residues identified	Plant/orphan receptors	Med high leve test ACG dire
15498549 CHASE structure prediction	Structural/ computational	Qualifies	CHASE fold & ligand pocket	CHASE resembles PAS/PYP-like small-molecule sensor domain	Bioinformatics	Med Pre leve
18216168 Cytokinins in Dictyostelium	Genetic/pathway	Qualifies	Which receptor senses	Cytokinins act via DhkB, not ACG	<i>Dictyostelium</i>	Hig CHA the

Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con- limi
			cytokinins in Dictyostelium?			orga hor rece
15590560 AC review	Review/database	Orientation	ACG's role among the 3 cyclases	ACG "acts as an osmosensor ... controlling spore germination"	<i>Dictyostelium</i>	Rev orie
10373524 DhkA/SDF-2	Direct (topology, enzymology)	Competing (identifies the real peptide receptor)	What receptor senses peptide SDF-2?	DhkA histidine kinase with extracellular ligand loop transduces SDF-2	<i>Dictyostelium</i>	Hig Gen pep rece Dhk ACC

GO Curation Implications

Lead (requires curator verification): Flag **GO:0001653 (peptide receptor activity)** on Q03101 for **removal or NOT-qualification**, or at minimum reclassify as **non-core, low-confidence propagated (IBA) annotation**.

- **Aspect:** Molecular Function.
- **Rationale:** The term is IBA-only and its `withFrom` provenance traces to natriuretic peptide receptors NPR1/NPR2. ACG's experimentally supported MF terms (adenylate cyclase, osmosensor) and its intramolecular, ligand-independent activation contradict a direct peptide-receptor function. This is family-level over-annotation driven by shared receptor-cyclase architecture and the PF00211 "Guanylate_cyc" catalytic-domain label.
- **Retain as core:** GO:0004016 (adenylate cyclase activity, IDA) and GO:0005034 (osmosensor activity, IDA).

- **Do not substitute "protein binding":** The informative, supported terms are the adenylate-cyclase and osmosensor activities; GO:0042802 (identical protein binding, IPI) captures dimerization but is not a substitute for core catalytic/sensory functions.
- **If NOT-qualification is preferred over deletion:** annotate GO:0001653 with a NOT qualifier and an ECO code reflecting curatorial review, citing the intramolecular osmosensor reconstitution ([PMID: 14718564](#)) and the absence of any peptide-binding assay.

Mechanistic Scope

The molecular function under test is **direct peptide receptor activity** — ACG binding an extracellular peptide ligand at its own extracellular domain and transducing that binding into a signaling output.

- **Direct gene-product activity (supported):** osmolality sensing (intramolecular) and ATP→cAMP catalysis — properties of the ACG polypeptide itself, demonstrated by heterologous reconstitution and enzyme assays.
- **Downstream / pathway consequences (not the MF being tested):** cAMP→PKA activation→inhibition of spore germination — a developmental output, not evidence of a receptor–peptide interaction.
- **Separate signaling modules (not ACG):** SDF-1 (intracellular PKA substrate), SDF-2 (peptide sensed by DhkA), GABA (via Gr1E GPCR), cytokinins (via DhkB). None implicates ACG as a peptide receptor.

There is no experimental measurement of ACG at the level required to support GO:0001653 — a binding assay or a functional readout demonstrating peptide-dependent activation of ACG.

Conflicts and Alternatives

1. **Paralog/family confusion (primary explanation).** ACG co-clusters in PANTHER PTN000229249 with mammalian natriuretic peptide receptors, which *are* peptide receptors; ACG is not. IBA cannot distinguish the divergence of the extracellular sensor (CHASE small-molecule/osmo-sensor in ACG vs. natriuretic-peptide-binding ectodomain in NPR1/2), so it over-projects the term.

2. **Architecture-driven false positive.** ACG's catalytic domain is annotated PF00211 ("Guanylate_cyc"), reinforcing the guanylyl-cyclase-receptor analogy even though ACG functions as an *adenylyl* cyclase — likely amplifying the mis-annotation.
 3. **Organism-specific divergence.** Dictyostelium uses histidine-kinase receptors (DhkA, DhkB) and GPCRs (GrIE) for peptide/hormone signaling — not receptor guanylyl/adenylyl cyclases. The mammalian receptor-cyclase paradigm does not map onto ACG's biology.
 4. **Alternative reading of the seed's "dependence" clause.** One could hypothesize SDF-1 signaling *depends on* ACG-generated cAMP. But SDF-1 acts in prespore cells during sporulation, whereas ACG functions in mature spores during germination; the stages and cell types differ, and no genetic epistasis links them. The dependence interpretation is unsupported.
 5. **What would genuinely conflict with the refutation:** a direct SDF-1 (or other peptide) binding assay on ACG, or an *acg*[−] phenotype in SDF-1-driven sporulation. Neither exists in the literature reviewed.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
No direct negative binding assay	Primary literature (osmosensor, SDF-1, SDF-2 papers)	"Refuted" rests on absence of positive evidence + provenance, not a published no-binding result	In-vitro peptide-binding assay (SPR/ITC) of ACG ectodomain vs. SDF-1/candidate peptides
SDF-1 receptor identity unknown	PMID: 9373946 , PMID: 9473320	If SDF-1's receptor were ever shown to be ACG, the hypothesis would revive	Identify SDF-1 surface receptor by crosslinking/genetics; test <i>acg</i> ⁻ SDF-1 response
CHASE ligand of ACG not directly assayed	CHASE family papers (PMID: 17439640 , 15498549)	ACG's CHASE is inferred as small-molecule/osmosensor by homology, not directly tested for ligands	Ligand screen of purified ACG CHASE domain
QuickGO provenance read manually	QuickGO/UniProt records	Confirms carry-over but is database-level; PANTHER membership should be re-verified at curation time	Curator re-checks PTN000229249 tree and current IBA propagation
Structural model of ACG ectodomain not built here	Domain annotations only (Pfam/InterPro)	A structure could confirm the absence of a peptide-binding groove	AlphaFold model + pocket analysis of ACG residues 86–317

Discriminating Tests

- 1. Direct peptide-binding assay on ACG.** Purify the ACG extracellular/CHASE domain (~86–317) and test binding to SDF-1 and other candidate peptides by SPR, ITC, or MST. *No binding* → refutes conclusively; *specific binding* → reopens the hypothesis.
- 2. Genetic epistasis of SDF-1 vs. ACG.** Test whether *acg*⁻ cells retain a normal SDF-1 sporulation response. If yes, SDF-1 signaling is ACG-independent.

3. **Yeast reconstitution + peptide challenge.** Extend the Saran & Schaap yeast system ([PMID: 14718564](#)) by adding candidate peptides to test for peptide-dependent ACG activation beyond osmotic stimulation.
4. **Structural pocket analysis.** Build an AlphaFold model of the ACG ectodomain and computationally dock SDF-1; compare its pocket geometry to the NPR1/NPR2 natriuretic-peptide-binding ectodomain to confirm fold divergence.
5. **PANTHER tree re-inspection.** Verify that GO:0001653 propagation in PTN000229249 originates solely from natriuretic peptide receptors and that no experimentally annotated non-mammalian member independently supports peptide binding.

Proposed Follow-up Experiments / Actions (Curation Leads)

All items are **leads requiring curator verification**.

Candidate action change: - Set **GO:0001653 (peptide receptor activity)** to **removed** or **NOT-qualified** for Q03101; document as IBA family carry-over from natriuretic peptide receptors. Retain GO:0004016 and GO:0005034 as core MF terms; retain GO:0042802 for dimerization.

Candidate references + snippets to verify: - [PMID: 14718564](#) — *"This strongly suggests that the ACG osmosensor is intramolecular, which would define ACG as the first characterized primary osmosensor in eukaryotes."* - [PMID: 8798577](#) — *"ACG is an osmosensor controlling spore germination through activation of protein kinase A."* - [PMID: 9373946](#) — *"the response of prespore cells to SDF involves an intracellular pathway dependent on PKA."* - [PMID: 9473320](#) — *tagC⁻/dhkA⁻/regA⁻ strains "all sporulated efficiently when SDF-1 was added."* - [PMID: 18216168](#) — *cytokinins act "through a different histidine kinase, DhkB."*

Candidate retained GO terms: - MF: GO:0004016 (adenylate cyclase activity) — retain, core. - MF: GO:0005034 (osmosensor activity) — retain, core. - MF: GO:0042802 (identical protein binding) — retain (dimerization). - BP: consider osmosensory-signaling / regulation of spore germination terms consistent with [PMID: 8798577](#).

Suggested curator questions: - Does any experimentally annotated member of PANTHER PTN000229249 support GO:0001653 independently of natriuretic peptide receptors? - Is there any post-2004 primary study assigning a peptide ligand to ACG? (None found here.)

Suggested experiments: Items 1–3 under *Discriminating Tests* (direct binding assay; *acg⁻* SDF-1 epistasis; yeast reconstitution + peptide challenge).

Notes on Reproducibility and Provenance

The conclusions rest on (a) the **evidence-code contrast** in UniProt Q03101 (IBA for GO:0001653 vs. IDA/IPI for characterized functions), (b) the **QuickGO** **withFrom** **provenance** listing NPR1/NPR2 as the IBA seed proteins, and (c) primary experimental papers on ACG osmosensing and SDF-1 signaling. Where a specific value (e.g., PANTHER family ID PTN000229249) is quoted, it should be re-verified by the curator at the current database version, since IBA propagations can change over time. No peptide-binding assay for ACG exists in the literature reviewed; the refutation is therefore an evidence-of-absence plus positive-provenance argument, stated conservatively.

Prepared for AI Gene Review — hypothesis-level curation support. All annotations flagged here are leads requiring curator verification.

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